LANDFORMS OF GLACIATION / HIMALAYAN GLACIERS-GLOF

GLACIER BUDGET → RETREAT PARADOX → EROSION SUITE → FJORD-RIA FIREWALL → DEPOSITIONAL SORTING → ICE-MASS FAMILY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00 — STAGE 00 GLACIER BUDGET

GLACIER BUDGET | SNOWFALL, AVALANCHE AND REFREEZING | ANNUAL GAIN EQUALS ANNUAL LOSS | MELT, SUBLIMATION AND CALVING WHERE RELEVANT | NET BALANCE | GEOMETRY RESPONDS WITH A LAG

DEFINITION / COMPONENTS
- ACCUMULATION → snowfall, avalanche and refreezing
- ELA → annual gain equals annual loss

TRIGGER / MECHANISM
- ABLATION → melt, sublimation and calving where relevant
- NET BALANCE → geometry responds with a lag

EFFECT / INTERVENTION
- ACCUMULATION → snowfall, avalanche and refreezing
- ELA → annual gain equals annual loss

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NET BALANCE → geometry responds with a lag.

01 — STAGE 01 RETREAT PARADOX

RETREAT PARADOX | ICE PARTICLES | CONTINUE MOVING DOWNSLOPE | RECEIVES FORWARD ICE FLUX | ABLATION DELIVERY | SNOUT SHIFTS UP-VALLEY | METRIC RULE | FLOW DIRECTION DIFFERS FROM MARGIN CHANGE

ICE PARTICLES → continue moving downslope → TERMINUS → receives forward ice flux → ABLATION DELIVERY → snout shifts up-valley → METRIC RULE → flow direction differs from margin change

DEFINITION / COMPONENTS
- ICE PARTICLES → continue moving downslope
- TERMINUS → receives forward ice flux

TRIGGER / MECHANISM
- ABLATION DELIVERY → snout shifts up-valley
- METRIC RULE → flow direction differs from margin change

EFFECT / INTERVENTION
- ICE PARTICLES → continue moving downslope
- TERMINUS → receives forward ice flux

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: METRIC RULE → flow direction differs from margin change.

02 — STAGE 02 EROSION SUITE

EROSION SUITE | JOINTED BLOCKS REMOVED | POLISH, STRIAE AND ROCK FLOUR | HEADWALL SYSTEM | HANGING VALLEY | VALLEY-SCALE OVERDEEPENING

PLUCKING → jointed blocks removed → ABRASION → polish, striae and rock flour → CIRQUE / ARETE / HORN → headwall system → U-TROUGH / HANGING VALLEY → valley-scale overdeepening

DEFINITION / COMPONENTS
- PLUCKING → jointed blocks removed
- ABRASION → polish, striae and rock flour

TRIGGER / MECHANISM
- CIRQUE / ARETE / HORN → headwall system
- U-TROUGH / HANGING VALLEY → valley-scale overdeepening

EFFECT / INTERVENTION
- PLUCKING → jointed blocks removed
- ABRASION → polish, striae and rock flour

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CIRQUE / ARETE / HORN → headwall system.

03 — STAGE 03 FJORD-RIA FIREWALL

FJORD-RIA FIREWALL | GLACIAL TROUGH LATER FLOODED BY SEA | RIVER VALLEY LATER DROWNED BY SEA | FJORD CUES | DEEP WATER, STEEP WALLS, HANGING VALLEYS | INUNDATION PRESERVES DIFFERENT PRE-EXISTING VALLEYS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
FJORD → glacial trough later flooded by sea	FJORD CUES → deep water, steep walls, hanging valleys	FJORD → glacial trough later flooded by sea
RIA → river valley later drowned by sea	RULE → inundation preserves different pre-existing valleys	RIA → river valley later drowned by sea

DEFINITION / COMPONENTS
- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

TRIGGER / MECHANISM
- FJORD CUES → deep water, steep walls, hanging valleys
- RULE → inundation preserves different pre-existing valleys

EFFECT / INTERVENTION
- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FJORD CUES → deep water, steep walls, hanging valleys.

03 STAGE 03 FJORD-RIA FIREWALL

FJORD-RIA FIREWALL · GLACIAL TROUGH LATER FLOODED BY SEA · RIVER VALLEY LATER DROWNED BY SEA · FJORD CUES · DEEP WATER, STEEP WALLS, HANGING VALLEYS · INUNDATION PRESERVES DIFFERENT PRE-EXISTING VALLEYS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
FJORD → glacial trough later flooded by sea	FJORD CUES → deep water, steep walls, hanging valleys	FJORD → glacial trough later flooded by sea
RIA → river valley later drowned by sea	RULE → inundation preserves different pre-existing valleys	RIA → river valley later drowned by sea

DEFINITION / COMPONENTS
- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

TRIGGER / MECHANISM
- FJORD CUES → deep water, steep walls, hanging valleys
- RULE → inundation preserves different pre-existing valleys

EFFECT / INTERVENTION
- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FJORD CUES → deep water, steep walls, hanging valleys.

04 STAGE 04 DEPOSITIONAL SORTING

DEPOSITIONAL SORTING · DIRECT ICE, GENERALLY UNSORTED · MELTWATER RIDGE IN OR UNDER ICE · BRAIDED MELTWATER, RELATIVELY SORTED · STREAMLINED DRIFT/TILL LANDFORM

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
TILL / MORAINE → direct ice, generally unsorted	OUTWASH → braided meltwater, relatively sorted	TILL / MORAINE → direct ice, generally unsorted
ESKER → meltwater ridge in or under ice	DRUMLIN → streamlined drift/till landform	ESKER → meltwater ridge in or under ice

DEFINITION / COMPONENTS
- TILL / MORAINE → direct ice, generally unsorted
- ESKER → meltwater ridge in or under ice

TRIGGER / MECHANISM
- OUTWASH → braided meltwater, relatively sorted
- DRUMLIN → streamlined drift/till landform

EFFECT / INTERVENTION
- TILL / MORAINE → direct ice, generally unsorted
- ESKER → meltwater ridge in or under ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESKER → meltwater ridge in or under ice.

05 STAGE 05 ICE-MASS FAMILY

ICE-MASS FAMILY · GLACIER, ICE CAP, ICE SHEET · SHELF, ICEBERG AND SEA ICE · SEA LEVEL · GROUNDED LOSS ADDS OCEAN MASS · SHELF LOSS CAN ACCELERATE INLAND DISCHARGE

GROUNDED → glacier, ice cap, ice sheet

FLOATING → shelf, iceberg and sea ice → SEA LEVEL → grounded loss adds ocean mass → BUTTRESSING → shelf loss can accelerate inland discharge

DEFINITION / COMPONENTS
- GROUNDED → glacier, ice cap, ice sheet
- FLOATING → shelf, iceberg and sea ice

TRIGGER / MECHANISM
- SEA LEVEL → grounded loss adds ocean mass
- BUTTRESSING → shelf loss can accelerate inland discharge

EFFECT / INTERVENTION
- GROUNDED → glacier, ice cap, ice sheet
- FLOATING → shelf, iceberg and sea ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUTTRESSING → shelf loss can accelerate inland discharge.

06 STAGE 06 HIMALAYAN MAP HOOKS

HIMALAYAN MAP HOOKS · BARA SHIGRI · CHENAB SYSTEM · NUBRA-SHYOK-INDUS SYSTEM

GANGOTRI → Bhagirathi

ZEMU → Teesta → BARA SHIGRI → Chenab system → SIACHEN → Nubra-Shyok-Indus system

SYSTEM / DEFINITION
- GANGOTRI → Bhagirathi
- ZEMU → Teesta

PROCESS / MECHANISM
- BARA SHIGRI → Chenab system
- SIACHEN → Nubra-Shyok-Indus system

SPATIAL / INDIA PATTERN
- GANGOTRI → Bhagirathi
- ZEMU → Teesta

HAZARD / LIMIT / RESPONSE
- BARA SHIGRI → Chenab system
- SIACHEN → Nubra-Shyok-Indus system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARA SHIGRI → Chenab system; SIACHEN → Nubra-Shyok-Indus system.

06 — STAGE 06 HIMALAYAN MAP HOOKS

HIMALAYAN MAP HOOKS · BARA SHIGRI · CHENAB SYSTEM · NUBRA-SHYOK-INDUS SYSTEM

GANGOTRI → Bhagirathi

ZEMU → Teesta → BARA SHIGRI → Chenab system → SIACHEN → Nubra-Shyok-Indus system

SYSTEM / DEFINITION
- GANGOTRI → Bhagirathi
- ZEMU → Teesta

PROCESS / MECHANISM
- BARA SHIGRI → Chenab system
- SIACHEN → Nubra-Shyok-Indus system

SPATIAL / INDIA PATTERN
- GANGOTRI → Bhagirathi
- ZEMU → Teesta

HAZARD / LIMIT / RESPONSE
- BARA SHIGRI → Chenab system
- SIACHEN → Nubra-Shyok-Indus system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARA SHIGRI → Chenab system; SIACHEN → Nubra-Shyok-Indus system.

07 — STAGE 07 WATER-TIMING SEQUENCE

WATER-TIMING SEQUENCE · EARLY LOSS · LOCAL MELT CONTRIBUTION MAY RISE · STORED ICE DECLINES · BUFFERING WEAKENS · LEAN SEASON · RELIABILITY RISK GROWS LOCALLY · MONSOON, SNOW AND GROUNDWATER VARY BY BASIN

EARLY LOSS → local melt contribution may rise → STORED ICE DECLINES → buffering weakens → LEAN SEASON → reliability risk grows locally → QUALIFIER → monsoon, snow and groundwater vary by basin

DEFINITION / COMPONENTS
- EARLY LOSS → local melt contribution may rise
- STORED ICE DECLINES → buffering weakens

TRIGGER / MECHANISM
- LEAN SEASON → reliability risk grows locally
- QUALIFIER → monsoon, snow and groundwater vary by basin

EFFECT / INTERVENTION
- EARLY LOSS → local melt contribution may rise
- STORED ICE DECLINES → buffering weakens

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFIER → monsoon, snow and groundwater vary by basin.

08 — STAGE 08 GLACIAL-LAKE TEST

GLACIAL-LAKE TEST · HOW WATER IS IMPOUNDED · SLOPE + ICE · POSSIBLE DISPLACEMENT TRIGGER · DAM CONDITION · OVERTOPPING, PIPING OR EROSION · EXPOSURE CONVERTS HAZARD INTO RISK

BASIN / DAM → how water is impounded

SLOPE + ICE → possible displacement trigger → DAM CONDITION → overtopping, piping or erosion → DOWNSTREAM → exposure converts hazard into risk

DEFINITION / COMPONENTS
- BASIN / DAM → how water is impounded
- SLOPE + ICE → possible displacement trigger

TRIGGER / MECHANISM
- DAM CONDITION → overtopping, piping or erosion
- DOWNSTREAM → exposure converts hazard into risk

EFFECT / INTERVENTION
- BASIN / DAM → how water is impounded
- SLOPE + ICE → possible displacement trigger

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BASIN / DAM → how water is impounded.

09 — STAGE 09 GLOF CASCADE

GLOF CASCADE · SLOPE FAILURE, INFLOW OR INSTABILITY · DAM EROSION BEGINS · SUDDEN RELEASE + SEDIMENT · CONSTRICTIONS AND ASSETS SHAPE CONSEQUENCES

TRIGGER → slope failure, inflow or instability → WAVE / OVERTOPPING → dam erosion begins → BREACH → sudden release + sediment → VALLEY → constrictions and assets shape consequences

DEFINITION / COMPONENTS
- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

TRIGGER / MECHANISM
- BREACH → sudden release + sediment
- VALLEY → constrictions and assets shape consequences

EFFECT / INTERVENTION
- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BREACH → sudden release + sediment; VALLEY → constrictions and assets shape consequences.

09 — STAGE 09 GLOF CASCADE

GLOF CASCADE | SLOPE FAILURE, INFLOW OR INSTABILITY | DAM EROSION BEGINS | SUDDEN RELEASE + SEDIMENT | CONSTRICTIONS AND ASSETS SHAPE CONSEQUENCES

TRIGGER → slope failure, inflow or instability → WAVE / OVERTOPPING → dam erosion begins → BREACH → sudden release + sediment → VALLEY → constrictions and assets shape consequences

DEFINITION / COMPONENTS

- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

TRIGGER / MECHANISM

- BREACH → sudden release + sediment
- VALLEY → constrictions and assets shape consequences

EFFECT / INTERVENTION

- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BREACH → sudden release + sediment; VALLEY → constrictions and assets shape consequences.

10 — STAGE 10 RISK-REDUCTION STACK

RISK-REDUCTION STACK | SATELLITE INVENTORY | CHANGING LAKES AND SLOPES | FIELD ASSESSMENT | DAM MATERIAL AND THRESHOLDS | WARNING + DRILLS | LAST-MILE ACTION | LAND USE + DESIGN

SATELLITE INVENTORY → changing lakes and slopes

FIELD ASSESSMENT → dam material and thresholds → WARNING + DRILLS → last-mile action → LAND USE + DESIGN → reduce exposure and fragility

DEFINITION / COMPONENTS

- SATELLITE INVENTORY → changing lakes and slopes
- FIELD ASSESSMENT → dam material and thresholds

TRIGGER / MECHANISM

- WARNING + DRILLS → last-mile action
- LAND USE + DESIGN → reduce exposure and fragility

EFFECT / INTERVENTION

- SATELLITE INVENTORY → changing lakes and slopes
- FIELD ASSESSMENT → dam material and thresholds

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LAND USE + DESIGN → reduce exposure and fragility.

11 — STAGE 11 CRYOSPHERE ANSWER SPINE

CRYOSPHERE ANSWER SPINE | GLACIER, LAKE, GLOF AND RISK | GLACIER-BASIN RELATION | BUDGET, TRIGGER AND BREACH | HETEROGENEITY, ATTRIBUTION AND DATA DATE

DEFINE → glacier, lake, GLOF and risk → MAP → glacier-basin relation → EXPLAIN → budget, trigger and breach → QUALIFY → heterogeneity, attribution and data date

DEFINITION / COMPONENTS

- DEFINE → glacier, lake, GLOF and risk
- MAP → glacier-basin relation

TRIGGER / MECHANISM

- EXPLAIN → budget, trigger and breach
- QUALIFY → heterogeneity, attribution and data date

EFFECT / INTERVENTION

- DEFINE → glacier, lake, GLOF and risk
- MAP → glacier-basin relation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLAIN → budget, trigger and breach; QUALIFY → heterogeneity, attribution and data date.

E — EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | ABOUT 15,000 GLACIERS IN THE HIMALAYA, BETWEEN THE TWO | LOWEST LEVEL OF PERPETUAL SNOW; IT VARIES WITH LATITUDE | GREATER SNOWFALL/ACCUMULATION CAN LOWER THE CLIMATIC SNOWLINE, WHILE ARIDITY | MAIN GLACIERS LIE IN THE GREATER AND TRANS-HIMALAYA (KARAKORAM | THE LESSER HIMALAYA HAS ONLY SMALL GLACIERS (TRACES IN | SIACHEN LIES IN THE EASTERN KARAKORAM UNDER INDIAN CONTROL. | SOUTH LHONAK IS A MORAINE-DAMMED GLACIAL LAKE THAT EXPANDED

OPTIONAL PROCESS DEPTH

- Khullar: about 15,000 glaciers in the Himalaya, between the two syntaxial bends (east & west).
- Snowline = lowest level of perpetual snow; it varies with latitude, precipitation and local topography.
- Greater snowfall/accumulation can lower the climatic snowline, while aridity raises it; aspect,
- Main glaciers lie in the Greater and Trans-Himalaya (Karakoram, Ladakh, Zaskar)

DATA / INSTITUTION LIMIT

- the Lesser Himalaya has only small glaciers (traces in Pir-Panjal, Dhauladhar).
- Siachen lies in the eastern Karakoram under Indian control. Biafo, Baltoro and Hispar are in the
- South Lhonak is a moraine-dammed glacial lake that expanded as its glacier retreated; use a named
- The October 2023 flood involved lake outburst/overtopping after intense precipitation and a

CONTEMPORARY PARALLEL

- Mitigation: glacial-lake monitoring, early-warning systems, and restricting development in the flood path (NDMA GLOF guidelines).
- Khullar's ~15,000 figure is a source-era Himalayan estimate; modern inventory counts are method-dependent.
- Snowline controls: latitude, precipitation, topography.
- More accumulation generally lowers snowline; exact north-south contrast varies by Himalayan sector.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels